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Dear Prof. Giulia Galli

April 28, 2025

Dear Prof. Giulia Galli,

I am writing to express my strong interest in joining your group at the University of Chicago as a visiting researcher. I am currently affiliated with the University of Tokyo and consider visiting your group under the formal dispatch of Prof. Synge Todo. I would be honored to pursue collaborative research under your supervision.

My research background lies in first-principles calculations, molecular dynamics, and machine-learning or data-driven modeling of dielectric and vibrational properties of condensed matter systems. During my PhD and subsequent research activities, I have developed two complementary methodological approaches to dielectric-property calculations. One is a first-principles framework for crystalline materials that combines density functional theory with anharmonic lattice-dynamics calculations, including phonon anharmonicity, and was implemented through extensions to the open-source code ALAMODE. The other is a first-principles and machine-learning-based approach for molecular liquids, in which machine-learning models of dipole moments are trained on *ab-initio* Wannier function data, developed as the open-source software MLWC. Through these projects, I have gained experience in developing, maintaining, and operating scalable computational software in collaborative environments.

I am particularly interested in exploring how first-principles and machine-learning approaches can be combined to describe electronic properties of complex materials at scale, and I believe that your group's work on electronic-structure theory and materials modeling provides an ideal environment for this research. I am confident that my background in computational materials science and software-oriented research would allow me to contribute productively to ongoing projects in your group.

Thank you very much for considering my application. I would be grateful for the opportunity to discuss possible research directions. Please find my short CV and publication list attached for your reference.

Yours faithfully,

Tomohito Amano

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Work and Research

2026/4–present **Project Researcher**, *Prof. Syngé Todo, The University of Tokyo, Tokyo*

Development of first-principles and machine-learning-based methodologies to compute spin relaxation time of crystals.

2024/4–2026/3 **Digital Consultant**, *PwC Consulting LLC, Tokyo*

Design and deployment of large-scale generative AI applications on cloud infrastructure, with a focus on reliability, scalability, and long-term maintainability.

Detailed achievements:

- Designed, developed, and maintained enterprise-grade generative AI applications deployed on Microsoft Azure, serving approximately 10,000 users.
- Experienced end-to-end system development, including application architecture design, cloud resource configuration, deployment pipelines, and operational monitoring.
- Contributed to a team-based development effort involving approximately 10 engineers, following Git-based workflows with code reviews and collaborative version control practices.

2024/4–2026/3 **Visiting Researcher**, *Prof. Shinji Tsuneyuki, The University of Tokyo, Tokyo*

Continued the research projects described in the PhD Course section, with a primary focus on finalizing data analysis, publishing results, and maintaining the developed computational frameworks.

2021/4–2024/3 **PhD Course**, *Prof. Shinji Tsuneyuki, The University of Tokyo, Tokyo*

Development of first-principles and machine-learning-based methodologies to compute dielectric properties of crystals, molecular liquids, and polymers.

Detailed achievements:

- Conducted joint research projects with Fujitsu Ltd. and JSR Corporation on dielectric materials for next-generation communication technologies.
- Developed a machine-learning-assisted methodology to evaluate dielectric properties of polymers, integrating first-principles simulations with data-driven modeling.
 - Performed large-scale first-principles simulations using the supercomputer Fugaku
 - Developed an open-source software package MLWC (Machine Learning Wannier Centers) for predicting dielectric properties of condensed molecular systems.
 - Developed a robust machine learning force field for liquid polymers using active sampling scheme, enabling long-time and large-scale simulations.
- Established an *ab-initio* framework for calculating dielectric functions of strongly anharmonic crystals beyond the harmonic approximation.
 - Extended the open-source lattice-dynamics code ALAMODE by implementing new functionalities for dielectric function and Green's function calculations based on anharmonic phonon theory.

2019/4–2021/3 **Master Course**, *Prof. Shinji Tsuneyuki, The University of Tokyo, Tokyo*

Numerical investigation of the Holstein model in the strong-coupling regime using coherent potential approximation, Monte Carlo simulations, and Green's function analysis.

2018/4–2018/9 **Bachelor Course**, *Prof. Hiroyoshi Sakurai, Prof. Kathrin Wimmer, The University of Tokyo, Tokyo*

Measurement of the intrinsic γ ray efficiency of GAGG(Ce) scintillators up to 10 MeV Benchmark experiment for γ ray scintillator GAGG at RIKEN. Designed/constructed measurement circuit.

Computer skills

Python	Proficient	<i>Applied to machine-learning model development, data analysis, and scientific post-processing.</i>
C/C++	Proficient	<i>Used for high performance calculations including Green's function calculations and anharmonic phonon calculations.</i>
Linux	intermediate	<i>Experienced in maintaining a laboratory computing cluster (~20 CPUs), including software installation, job management, and network configuration.</i>
fortran	basic	<i>Capable of reading and modifying open-source DFT codes such as Quantum ESPRESSO.</i>

Software

VASP	Proficient	<i>DFT calculations for lattice dynamics and anharmonic phonon properties, including post-processing force-constant extraction and ab-initio molecular dynamics calculations.</i>
Quantum Espresso	Experienced	<i>DFT calculations for lattice dynamics and anharmonic phonon properties, including post-processing force-constant extraction.</i>
CPMD	Proficient	<i>ab-initio molecular dynamics calculations and Wannier functions calculations, with development of custom post-processing pipelines to extract machine-learning-ready datasets.</i>
LAMMPS	Intermediate	<i>Classical and machine-learning molecular dynamics simulations, including large-scale trajectory generation and analysis for molecular systems.</i>
DeepMD	Intermediate	<i>Development and application of deep neural network interatomic potentials trained on ab-initio data, and their integration with LAMMPS for large-scale molecular dynamics simulations.</i>
ALAMODE	Proficient	<i>Extended the ALAMODE code by implementing routines for dielectric function and Green's function calculations</i>
MLWC	Author	<i>Open-source software for constructing machine learning model for dipole moment applicable to molecular systems, using ab-initio Wannier functions as training data.</i>

Award

- 2024/3 **The School of Science Encouragement Award**
 Awarded for outstanding academic achievements and research performance in the School of Science, the University of Tokyo.
- 2024/3 **Springer Thesis Award**
 Selected for the Springer Thesis Award for an excellent doctoral thesis in condensed matter physics.

Scholarship

- 2019/12–2024/3 **Material Education program for the future leaders in Research, Industry, and Teaching. MERIT**
 Competitive scholarship awarded to outstanding students in materials science.

Research Interests

<i>ab-initio</i> calculation	First-principles methods for electronic, vibrational, and dielectric properties of condensed matter and molecular systems.
Molecular dynamics	Classical, <i>ab-initio</i> , and machine learning molecular dynamics simulations for crystalline and liquid systems.
Phonon Anharmonicity	Anharmonic lattice dynamics and phonon-mediated phenomena beyond the harmonic approximation.
Machine learning	Machine learning techniques for modeling physical properties from first-principles data and enabling scalable materials simulations.

Publication

Refereed Papers

- T. Amano, T. Yamazaki, N. Matsumura, Y. Yoshimoto, S. Tsuneyuki, Transferability of the chemical bond-based machine learning model for dipole moment: the GHz to THz dielectric properties of liquid propylene glycol and polypropylene glycol, Phys. Rev. B 111, 165149 (2025). [\[Link\]](#) [\[Link\]](#)
- N. Matsumura, Y. Yoshimoto, T. Yamazaki, T. Amano, T. Noda, N. Ebata, T. Kasano, Y. Sakai, Generator of Neural Network Potential for Molecular Dynamics: Constructing Robust and Accurate Potentials with Active Learning for Nanosecond-Scale Simulations, J. Chem. Theory Comput. 21, 3832 (2025). [\[Link\]](#) [\[Link\]](#)
- T. Amano, T. Yamazaki, and S. Tsuneyuki, Chemical bond based machine learning model for dipole moment: Application to dielectric properties of liquid methanol and ethanol, Phys. Rev. B 110, 165159 (2024). [\[Link\]](#)[\[arXiv\]](#)
- T. Amano, T. Yamazaki, R. Akashi, T. Tadano, and S. Tsuneyuki, Lattice Dielectric Properties of Rutile TiO₂: First-Principles Anharmonic Self-Consistent Phonon Study, Phys. Rev. B 107, 094305 (2023). [\[Link\]](#) [\[arXiv\]](#)

Conference Proceedings

- T. Amano, T. Yamazaki, and S. Tsuneyuki, The first-principles study of THz dielectric properties with a machine learning model for dipole moment, accepted for Proceedings of 34rd IUPAP Conference on Computational Physics (refereed)

Book

- T. Amano, First-Principles and Machine Learning Study of Anharmonic Vibration and Dielectric Properties of Materials, Springer Thesis (2025). [\[Link\]](#)

Oral Presentation

domestic

- T. Amano "First-principles and machine learning study of anharmonic vibration and dielectric properties of materials" Theory Seminar at the Institute for Solid State Physics, Chiba, Japan, Jan. 26, 2024 (invited)
- T. Amano, T. Yamazaki, and S. Tsuneyuki "Dielectric properties of polymers by machine-learning local dipole moments" The Physical Society of Japan (JPS) 2023 Autumn Meeting, Sendai, Japan, Sep. 17, 2023
- T. Amano, T. Yamazaki, and S. Tsuneyuki "First-principles study of THz dielectric properties using a machine learning model for dipole moment" The Physical Society of Japan (JPS) 2023 Spring Meeting, online, Mar. 22, 2023
- T. Amano, T. Yamazaki, R. Akashi, T. Tadano, S. Tsuneyuki "First-principles study of lattice dielectric function in rutile TiO₂" YITP Workshop Fundamentals in density functional theory (DFT2022), Kyoto, Japan, Dec. 12, 2022
- T. Amano, T. Yamazaki, R. Akashi, T. Tadano, and S. Tsuneyuki "First-principles study of lattice dielectric function in rutile TiO₂" The Physical Society of Japan (JPS) 2022 Autumn Meeting, Tokyo, Japan, Sep. 14, 2022

international

- T. Amano, T. Yamazaki, S. Tsuneyuki "The first-principles study of THz dielectric properties with a machine learning model for dipole moment" Advanced Materials Research Grand Meeting MRM2023, Kyoto, Japan, Dec. 12, 2023

- T. Amano, T. Yamazaki, S. Tsuneyuki "First-principles study of THz dielectric properties of liquid molecules with a machine learning model for dipole moments" American Physics Society April Meeting 2023, Minneapolis, USA, Jul. 7, 2023
- T. Amano, T. Yamazaki, R. Akashi, S. Tsuneyuki "The first-principles study of THz dielectric properties with a machine learning model for dipole moment" International Workshop on Theory of Many-Body Problems: Hierarchical Structure and Machine Learning (HISML) 2023, Chiba, Japan, Oct. 10, 2023
- T. Amano, T. Yamazaki, R. Akashi, S. Tsuneyuki "First-principles study of lattice dielectric function in rutile TiO_2 " 33rd IUPAP Conference on Computational Physics, Aug. 04, 2022

Poster presentations

domestic

- T. Amano, T. Yamazaki, S. Tsuneyuki "The machine learning simulation of THz dielectric properties(hp230124)" The 11th Project Report Meeting of the HPCI System Including Fugaku, Oct. 25, 2024
- T. Amano, T. Yamazaki, S. Tsuneyuki "The first-principles study of THz dielectric properties with a machine learning model for dipole moment" ipi Daikin symposium2023, Oct. 24, 2023
- T. Amano, T. Yamazaki, R. Akashi, T. Tadano, S. Tsuneyuki "Lattice Dielectric Properties of Rutile TiO_2 : First-Principles Anharmonic Self-Consistent Phonon Study" Frontiers of Computational Materials Science2023, Chiba, Japan, Apr. 03, 2023
- T. Amano, T. Yamazaki, S. Tsuneyuki "The first-principles study of THz dielectric properties of liquid molecules with a machine learning model for dipole moment" Spring School on Computational Physics2023, Mar. 13, 2023
- T. Amano, R. Akashi, S. Tsuneyuki "First-principles study of lattice dielectric function in rutile TiO_2 " Frontiers of Computational Materials Science2022, Chiba, Japan, May, 12, 2022

international

- T. Amano, T. Yamazaki, S. Tsuneyuki "First-principles study of THz dielectric properties of molecular liquids with a machine learning model for dipole moments" Psi-k 2025, Lausanne, Switzerland, Aug. 27, 2025
- T. Amano, T. Yamazaki, S. Tsuneyuki "The first-principles study of THz dielectric properties with a machine learning model for dipole moment" 34rd IUPAP Conference on Computational Physics, Kobe, Japan, Aug. 5, 2023
- T. Amano, N. Ogawa, W. Yamada, T. Ikeda, et al "Efficiency measurement of next generation scintillator GAGG at high energy using the 992KeV resonance of the $^{27}\text{Al}(p\&\gamma)^{28}\text{Si}$ reaction" 5th Joint Meeting of the APS Division of Nuclear Physics and the Physical Society of Japan, Hawaii, USA, Oct. 26, 2018